Chapter 1.

1.1. $-V_1+3V_2-2V_3 = -\begin{bmatrix}4\\0\\-2\\1\end{bmatrix}+3\begin{bmatrix}2\\-2\\8\\-1\end{bmatrix}-2\begin{bmatrix}-3\\1\\0\\6\end{bmatrix}$

$$= \begin{bmatrix}-4+6+6\\0-6-2\\2+24-0\\-1-3-12\end{bmatrix} = \begin{bmatrix}8\\-8\\26\\16\end{bmatrix}$$

1.2. (a) $AV = \begin{bmatrix}-7\\-9\end{bmatrix}$

$wB = \begin{bmatrix}-19 & 18 & -10\end{bmatrix}$

$DV = \begin{bmatrix}18\\-5\\-14\end{bmatrix}$

$V^tD = \begin{bmatrix}14 & 15 & -22\end{bmatrix}$

$Aw^t = \begin{bmatrix}-1 & 2 & -3\\4 & 1 & 1\end{bmatrix}\begin{bmatrix}-3\\5\end{bmatrix} \rightarrow$ you can't multiply a 2×3 matrix and a 2×1 matrix.

$Dc^t = \begin{bmatrix}8 & -5\\5 & -4\\-20 & 9\end{bmatrix}$

(b) $2A-3B = \begin{bmatrix}-11 & 7 & -6\\14 & -11 & 8\end{bmatrix}$

(c) $AD = \begin{bmatrix}-16 & -2 & -8\\6 & -7 & 22\end{bmatrix}$

DB is not defined since D has 3 columns and B only has 2 rows.

(d) $A+B = B+A = \begin{bmatrix}2 & 1 & -3\\2 & 2 & -1\end{bmatrix}$

(e) $(A+B)+C = A+(B+C) = \begin{bmatrix}-3 & 2 & -1\\6 & -3 & -4\end{bmatrix}$

(f) $2(A+B) = 2A+2B = \begin{bmatrix}4 & 2 & -6\\4 & 4 & -2\end{bmatrix}$

(g) $(3B)^t = 3B^t = \begin{bmatrix}9 & -6\\-3 & 9\\0 & -6\end{bmatrix}$

(h) $(A+B)^t = A^t+B^t = \begin{bmatrix}2 & 2\\1 & 2\\-3 & -1\end{bmatrix}$

1.3. Av will be an $m \times 1$ matrix. (i.e, a column m-vector).

1.4. $v^t w = [1\ 2\ 3]\begin{bmatrix}4\\5\\6\end{bmatrix} = [32]$

$vw^t = \begin{bmatrix}1\\2\\3\end{bmatrix}[4\ 5\ 6] = \begin{bmatrix}4 & 5 & 6\\ 8 & 10 & 12\\ 12 & 15 & 18\end{bmatrix}$

1.5 Proof: Let $v = \begin{bmatrix}v_1\\v_2\\\vdots\\v_n\end{bmatrix}$

$$\left.\begin{aligned} \alpha v &= \alpha\begin{bmatrix}v_1\\v_2\\\vdots\\v_n\end{bmatrix} = \begin{bmatrix}\alpha v_1\\\alpha v_2\\\vdots\\\alpha v_n\end{bmatrix} \leftarrow \text{scalar multiple} \\ v[\alpha] &= \begin{bmatrix}v_1\\v_2\\\vdots\\v_n\end{bmatrix}[\alpha] = \begin{bmatrix}v_1\alpha\\v_2\alpha\\\vdots\\v_n\alpha\end{bmatrix} \rightarrow \text{matrix multiplication.} \end{aligned}\right\} \Rightarrow \alpha v = v[\alpha]$$

Chapter 2.

2.1. (a) $\begin{bmatrix}1 & 2 & -3 & 7\\ -2 & -1 & 4 & 5\\ 3 & 3 & -6 & 4\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}0\\1\\3\end{bmatrix}$

2.2. $x_1 + 2x_2 - x_3 = 1$

$-2x_1 + 3x_2 + 5x_3 = -1$

$2x_1 + 3x_2 - 4x_3 = 2.$

2.3 There are many possibilities. For example, Let A be a 2×3 matrix and B be a 3×1 matrix, Then AB will be a 2×1 matrix, but BA is not defined.

2.4. (a) $(AB)C = A(BC) = \begin{bmatrix}-2 & 4\\ -2 & 4\end{bmatrix}$

(b) $A(B+C) = AB + AC = \begin{bmatrix}1 & 0\\ 5 & -4\end{bmatrix}$

(c) $(B+C)A = BA + CA = \begin{bmatrix}-9 & -10\\ 5 & 6\end{bmatrix}$

(d) $(3A)B = A(3B) = 3(AB) = \begin{bmatrix}-3 & 3\\ -3 & 3\end{bmatrix}$

(e) $(AB)^t = \begin{bmatrix}-1 & -1\\ 1 & 1\end{bmatrix}$, $A^tB^t = \begin{bmatrix}-2 & 2\\ -2 & 2\end{bmatrix} \neq (AB)^t$. $B^tA^t = \begin{bmatrix}-1 & -1\\ 1 & 1\end{bmatrix} = (AB)^t$

2.5 (a) $A^2 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$

$A^3 = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$

(b) $B^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ $B^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 8 \end{bmatrix}$

2.6. There are many solutions.

for example: $A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Then $AB = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = O_{2\times 2}$.

2.7. $(AB)^2 = A\underline{BA}B$ and $A^2B^2 = A\underline{AB}B$.

If $BA = AB$, then $A(BA)B = A(AB)B$, i.e., $(AB)^2 = A^2B^2$

So we need to choose A and B such that at least $AB \neq BA$.

There are many solutions.

for example:

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

Then $(AB)^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \neq A^2B^2 = \begin{bmatrix} -6 & 6 \\ -14 & 14 \end{bmatrix}$

2.8: By the analysis of 2.7, if $AB = BA$, then $(AB)^2 = A^2B^2$.

But there are times when $AB \neq BA$ and we still have $(AB)^2 = A^2B^2$

2.9. Proof: let $A = (a_{ij})_{m\times n}$, $B = (b_{ij})_{n\times k}$, then

$A^t = (\tilde{a}_{ij})_{n\times m}$, $B^t = (\tilde{b}_{ij})_{k\times n}$ where $\tilde{a}_{ij} = a_{ji}$, $\tilde{b}_{ij} = a_{ji}$

The (i,j)-component of $(AB)^t$

= The (j,i)-component of AB

$= \sum_{k=1}^{n} a_{jk} b_{ki} = \sum_{k=1}^{n} \tilde{a}_{kj} \tilde{b}_{ik} = \sum_{k=1}^{n} \tilde{b}_{ik} \tilde{a}_{kj}$

= The (i,j)-component of B^tA^t

$\Rightarrow (AB)^t = B^tA^t$